

THE IMPACT OF NUTRITION ON SELECTED ASPECTS OF DOMESTIC CAT BEHAVIOR – A LITERATURE REVIEW

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ABSTRACT

Nutrition of the domestic cat (*Felis catus*) affects not only physical health but also behavior and psychological well-being. The aim of this review was to synthesize current knowledge on the relationship between diet and feline behavior, with particular emphasis on physiological mechanisms, the role of the gut–brain axis, neurobiochemical foundations of behavioral regulation, and practical implications. Based on the literature analysis, it was demonstrated that the evolutionary adaptations of cats as obligate carnivores significantly influence their feeding patterns, and ad libitum feeding disrupts natural foraging rhythms and promotes the development of stereotypic behaviors. The gut–brain axis serves as a key mediator of diet's influence on behavior through microbiome modulation and the effects of nutraceuticals on inflammatory processes and

cognitive functions. Diet supplies neurotransmitter precursors (tryptophan, tyrosine, choline) and cofactors for their synthesis (B vitamins, magnesium, zinc), as well as modulates hormonal profiles involved in appetite and satiety regulation (ghrelin, leptin, PYY, CCK). The importance of palatability and food preferences in the context of food aversion, as well as the benefits of environmental enrichment through food puzzles, were highlighted. Based on the review, practical recommendations were formulated regarding meal fractionation, supplementation, environmental enrichment, and individualization of nutrition.

Keywords:

domestic cat, behavior, nutrition, gut–brain axis, gut microbiome, animal welfare

1. INTRODUCTION

In recent years, there has been significant progress in understanding the relationship between nutrition and the welfare of the domestic cat (*Felis catus*) and its behavior (12). The cat, as an obligate carnivore, has developed a range of unique metabolic and behavioral adaptations that are often insufficiently considered in domestic settings (11, 16). Contemporary research increasingly indicates that diet and its method of provision constitute key elements not only of physical health but also of the cat's psychological well-being.

The gut–brain axis has become a significant area of research, and its role in regulating behaviors related to stress, anxiety, and cognitive functions is currently the subject of intensive analysis. As Nicotra et al. (12) emphasize, nutraceuticals such as omega-3 fatty acids, prebiotics, probiotics, and plant extracts exhibit anti-inflammatory, immunomodulatory, and cognition-affecting properties, which have a direct impact on animal behavior (12). At the same time, McGrath et al. (11) point out that research on cognitive processes in cats in a nutritional context is still in its early stages, although a growing body of evidence indicates the key role of diet in maintaining cognitive functions, particularly in older animals.

The aim of this article is to present a comprehensive review of the latest literature concerning the impact of nutrition on cat behavior, with particular emphasis on physiological mechanisms, the gut–brain axis, neurobiochemical foundations of behavior, and practical implications for caregivers.

2. NATURAL FEEDING PATTERNS AND BEHAVIOR – EVOLUTIONARY FOUNDATIONS

Understanding the feeding behavior of the cat requires consideration of its evolutionary past. Members of the family Felidae are obligate carnivores, which is reflected in their anatomy, physiology, and behavior (11). The diet of wild cats consists almost exclusively of animal tissue, with birds and small mammals being the most common prey. Due to the small size of their prey, cats in the natural environment consume approximately 8–16 small meals per day (11, 17).

Metabolic adaptations of the cat to a diet based on animal protein and fat include an increased requirement for protein (two to three times higher than in omnivores) and for specific amino acids such as taurine, arginine, methionine, and cysteine, which cats cannot synthesize in sufficient quantities (11). Taurine deficiency can lead to blindness, reproductive problems, or dilated cardiomyopathy; arginine deficiency causes neurological disorders (11).

Watson et al. (17) note that free access to high-calorie dry food (ad libitum) disrupts the natural hunting-foraging rhythm, leading to boredom, frustration, and the development of stereotypical behaviors (excessive grooming, fabric sucking or chewing, pacing along walls, chewing on objects). Disrupted rhythm can result in rushed eating, a lack of natural foraging behaviors, and the shifting of activity to inappropriate times.

Research on feral cat populations indicates that aggressive behaviors and avoidance of human contact may be partially conditioned by the composition of the gut micro-

biome. In the work of Aizpurua et al. (1), it was demonstrated that feral cats are characterized by an increased capacity of the microbiome to break down amino acids and lipids, corresponding to their diet based on raw protein and fat. A positive association was observed between the microbial production of short-chain fatty acids, neurotransmitters, and vitamins and the intensity of aggressiveness. This discovery sheds new light on the mechanisms through which diet can influence behavior the gut–brain axis constitutes a bidirectional connection in which the gut microbiota modulates the functions of the central nervous system.

3. THE GUT–BRAIN AXIS AS A MEDIATOR OF DIET'S INFLUENCE ON BEHAVIOR

The gut–brain axis constitutes a bidirectional communication system between the gastrointestinal tract and the central nervous system, which plays a key role in the regulation of behavior, mood, and cognitive functions in animals (12). Imbalances in the gut microbiota (dysbiosis) can lead to low-grade inflammation, which affects brain function and may manifest as changes in behavior, such as increased anxiety or aggression.

3.1. Gut Microbiome and the Mental Health of Cats

Nicotra et al. (2025) indicate that omega-3 fatty acids, prebiotics, probiotics, and plant extracts exhibit anti-inflammatory and immunomodulatory properties that can indirectly influence behavior through the reduction of inflammation in the central nervous system. Particularly significant is the confirmation of the role of the gut–brain axis in regulating behaviors related to stress and anxiety, which opens new possibilities for nutritional interventions in the therapy of behavioral problems.

Research on the use of antimicrobial peptides (AMPs) derived from *Saccharomyces cerevisiae* provides significant evidence for the impact of dietary components on the stress response of cats through the modulation of gut microbiota. In a randomized study, Chen et al. (4) subjected 12 British Shorthair cats to two hours of transport and then placed them in a new environment. Cats receiving AMP supplementation showed significantly lower stress indicators during the first 3 days in the new environment, reduced activity on the day of transport, and extended nighttime sleep duration.

Biochemical analyses revealed a reduction in corticotropin-releasing hormone (CRH), cortisol (COR), serum amyloid A (SAA), and pro-inflammatory cytokines IL-1 β and IL-6, with a simultaneous increase in immunoglobulin G and Apo-A1. AMPs modulated the composition of the gut microbiome, increasing the abundance of *Bacteroides*, *Prevotella*, and *Collinsella* (4). This study demonstrates for the first time in such a comprehensive manner that nutritional intervention can alleviate stress through the reduction of inflammation, oxidative stress, and modulation of the gut–brain axis.

3.2. Nutraceuticals as a Tool Supporting Psychological Well-being

Nicotra et al. (12) indicate that, in addition to their direct effect on inflammation, nutraceuticals can support the human–animal bond through stress reduction and improved socialization. At the same time, the authors draw attention to the potential risks associated with improper use of nutraceuticals, which points to the need for further clinical research.

4. NEUROBIOCHEMICAL FOUNDATIONS OF NUTRITION'S IMPACT ON CAT BEHAVIOR – NUTRIENTS, NEUROTRANSMITTERS, AND HORMONES

Nutrients supplied through food constitute precursors of neurotransmitters, cofactors for enzymes synthesizing these compounds, and modulators of neuronal receptor activity (11, 12).

4.1. Tryptophan – Serotonin: Regulation of Anxiety, Impulsivity, and Aggression

Tryptophan is an exogenous amino acid that acts as a precursor to serotonin (through the intermediate metabolite 5-hydroxytryptophan) – a neurotransmitter that plays a key role in regulating mood, anxiety, impulsivity, and aggressive behaviors (11, 12). The availability of tryptophan in the central nervous system depends on its plasma concentration and its competition with other large neutral amino acids for transport across the blood–brain barrier (11).

4.2. Tyrosine – Dopamine and Norepinephrine: Motivation, Activity, and Alertness

Tyrosine serves as a precursor for catecholamines: dopamine, norepinephrine, and epinephrine. Dopamine plays a key role in the reward system and motivation, while norepinephrine is associated with wakefulness, arousal, and the stress response (11, 12).

Riva et al. (14) demonstrated that higher levels of dopamine and norepinephrine in plasma correlated with the absence of aggressive behaviors, suggesting that adequate availability of catecholamine precursors may support the proper functioning of the reward system and emotional regulation.

4.3. Glutamine, Glutamate, and GABA: Excitatory–Inhibitory Balance

Glutamate is the primary excitatory neurotransmitter, while GABA serves as the main inhibitory neurotransmitter. The balance between these systems is crucial for proper brain function, and its disruption can lead to hyperactivity, anxiety, and in extreme cases, seizures. Research on penicillin-induced epilepsy in cats showed that as the seizure state progresses, there is a loss of taurine and glutamate from tissues, which may contribute to increasing hyperexcitability (9).

4.4. Choline – Acetylcholine: Cognitive Functions and Attention

Choline is a precursor of acetylcholine—a neurotransmitter crucial for cognitive functions, attention, and memory. Diet can influence the functioning of the cholinergic system by providing choline (present in egg yolks, liver, and meat) and B vitamins, which are cofactors in acetylcholine metabolism. Choline deficiency can lead to cognitive dysfunction, impaired attention, and learning problems (11).

4.5. Taurine: CNS Development, Retinal and Brain Functions

Taurine is an amino acid that plays a particularly important role in the development of the central nervous system in cats. Cats have a limited capacity for taurine synthesis, making adequate dietary provision crucial. Taurine acts as a modulator of neuronal activity, a stabilizer of cell membranes, an antioxidant, and a regulator of calcium homeostasis.

Saransaari and Oja (15) demonstrated that taurine transport processes in the cat brain are highly efficient, especially during the developmental period. Potassium ion-stimulated taurine efflux from brain slices was several times higher in kittens than in adult cats. Taurine deficiency in kittens causes severe dysfunction in the cerebellar cortex and visual cortex. Taurine is essential for proper retinal development, and its deficiency leads to retinal degeneration and blindness (11, 15).

4.6. B Vitamins as Cofactors in Neurotransmitter Synthesis

B vitamins play a key role as cofactors in enzymatic reactions related to neurotransmitter metabolism:

Vitamin B6 (pyridoxine) is an essential cofactor for aromatic amino acid decarboxylase (AADC), converting 5-HTP to serotonin and L-DOPA to dopamine; it is also a cofactor for GAD, the enzyme synthesizing GABA.

Vitamin B9 (folic acid) and B12 (cobalamin) participate in the synthesis of S-adenosylmethionine (SAM) and in the remethylation of homocysteine to methionine, which is important for the synthesis of choline and acetylcholine.

Vitamin B3 (niacin) is a precursor of NAD and NADP; tryptophan can be converted to niacin, although this pathway is limited in cats (11).

Vitamin B5 (pantothenic acid) is a component of coenzyme A, essential in the synthesis of acetylcholine.

Deficiency of B vitamins can lead to behavioral disorders, including increased anxiety, aggression, and cognitive impairment (11).

4.7. Magnesium and Zinc as Modulators of NMDA Receptors

Magnesium acts as an endogenous blocker of the NMDA receptor ion channel under resting conditions, preventing calcium influx into the neuron. Magnesium deficiency can lead to excessive NMDA receptor activation, excitotoxicity, and cognitive impairment (11).

Zinc is an allosteric modulator of NMDA receptors and a cofactor for many enzymes related to neurotransmitter metabolism, including serotonin and dopamine synthesis (11).

4.8. Hormonal Regulation of Food Motivation, Exploratory Behaviors, and the Reward System

4.8.1. Ghrelin – The Hunger Hormone, Leptin – The Satiety Hormone

Ghrelin is a hormone predominantly secreted by gastric cells, which signals hunger and stimulates foraging behavior. Leptin, by contrast, is an anorexigenic hormone secreted by adipocytes, inhibiting appetite and increasing energy expenditure. A study by Akgün and Şahinduran (2) reported an unusual pattern in obese cats, characterized by elevated ghrelin and decreased leptin concentrations. Nevertheless, these findings are inconsistent with the predominant body of evidence, which indicates hyperleptinemia with concomitant leptin resistance, along with decreased or unaltered ghrelin levels. Consequently, these results should not be overgeneralized, and the observed phenomenon warrants further investigation.

4.8.2. Peptide YY (PYY) – The Satiety Hormone

PYY is secreted by intestinal L-cells in response to food intake, and its concentration increases particularly after protein-rich meals, contributing to appetite suppression. Studies in mice have shown that long-term consumption of a high-protein diet increases plasma PYY concentrations, reduces food intake, and limits weight gain and fat accumulation (3).

4.8.3. Cholecystokinin (CCK) – Satiety and Activity Regulation

In cats, as in other species, CCK is secreted in response to fat and protein intake, inhibiting gastric emptying and inducing satiety. A high-fat diet can lead to increased CCK secretion, which in humans is associated with postprandial drowsiness and reduced alertness. Although direct studies on this phenomenon in cats are limited, similarities in digestive system physiology suggest that an analogous mechanism may also occur in this species (18).

5. DIET COMPOSITION, INFLAMMATION, AND STRESS

Stress in cats involves the activation of the hypothalamic–pituitary–adrenal (HPA) axis and the sympathetic nervous system (4). Chronic stress leads to elevated cortisol levels, inflammation, and metabolic disorders that may manifest as behavioral problems.

Udell et al. (16) emphasize that the feeding approach should take into account not only the nutritional value of the diet but also its impact on satiety, weight control, and psychological well-being (16). A highly processed diet, poor in natural bioactive components, may contribute to chronic low-grade inflammation that negatively affects brain function and behavior. McGrath et al. (11) emphasize that age-related cognitive and behavioral changes in cats are often mild and mistakenly considered a natural

consequence of aging, whereas proper nutrition can delay these processes (11).

6. PALATABILITY AND FOOD PREFERENCES VS. BEHAVIOR

Cats are exceptionally picky eaters, which results from a limited number of taste buds (approximately 470 compared to approximately 1700 in dogs) and specific taste preferences. As described by Pekel et al. (13), cats lack the ability to perceive sweet taste due to a mutation in the gene encoding the sweet taste receptor (Tas1r2). Instead, they prefer umami tastes (corresponding to the presence of amino acids), which indicate that this taste adaptation may allow cats to perceive the taste of meat more sensitively. Cats are insensitive to salts and do not show a behaviorally significant reaction to sugars, which confirms their evolutionary specialization as carnivores (11).

Crucially from a behavioral perspective, negative feeding experiences in early life (neophobia) and dietary monotony can lead to persistent food aversion, manifesting as food refusal, weight loss, and in extreme cases, the development of compulsive foraging-related behaviors.

Watson et al. (17) emphasize that traditional methods of assessing palatability, based solely on measuring the amount of food consumed, are insufficient for fully understanding preferences. Analysis of behavioral indicators of acceptance, such as facial expressions, approach time to the bowl, and the sequence of foraging behaviors, is essential (17).

7. ENVIRONMENTAL ENRICHMENT THROUGH NUTRITION

The introduction of food puzzles constitutes one of the most effective methods of environmental enrichment for cats living in confinement. Ellis et al. (10), in the AAFP and ISFM guidelines, identify the provision of opportunities to perform natural foraging behaviors as one of the key pillars of environmental enrichment (10).

Dantas-Divers et al. (6) observed that 23 of 27 cats in a stable group interacted with the puzzle feeder, and the presence of the device did not affect the frequency of agonistic behaviors (6). Dantas et al. (5) emphasize that appropriately selected and gradually introduced food puzzles can benefit almost every cat, improving its psychological and physical well-being (5).

Delgado et al. (7) revealed an interesting paradox: unlike most species that prefer to work for food (contrafreeloading), cats show a clear preference for a free meal. The authors suggest that the lack of preference may result from the cat's evolutionary hunting strategy, but emphasize that the lack of preference does not mean a lack of benefits—food puzzles can support weight control, anxiety reduction, and improved well-being (7, 8).

8. SUMMARY AND PRACTICAL RECOMMENDATIONS

The latest literature unequivocally indicates that feeding the domestic cat extends far beyond the issue of balancing nutrients. The impact on behavior is conditioned by several interconnected mechanisms: interaction with the gut–brain axis and microbiome modulation, provision of neurotransmitter precursors and cofactors for their synthesis, hormonal regulation (ghrelin, leptin, PYY, CCK), and the degree to which the method of food presentation mimics the natural hunting sequence.

Based on the review of studies, the following practical recommendations can be formulated:

Regularity and meal fractionation: Avoid ad libitum feeding in favor of 3–5 small meals per day, which synchronizes the circadian rhythm and reduces frustration associated with waiting for food.

Optimization of the tryptophan-to-LNAA ratio: A diet with moderate protein content and an appropriate amount of carbohydrates can increase the tryptophan-to-LNAA ratio, facilitating serotonin synthesis and a calming effect. This may be beneficial for cats with anxiety and aggressive behaviors; however, the risk of obesity requires caloric control.

Ensuring adequate provision of catecholamine precursors: A diet rich in animal protein, providing appropriate amounts of tyrosine, supports the synthesis of dopamine and norepinephrine, which supports motivation, alertness, and the functioning of the reward system.

Taurine supplementation: Due to the key role of taurine in CNS development and limited synthesis capacity, it is essential to ensure adequate dietary taurine provision, especially during growth and in cats fed homemade diets.

Inclusion of B vitamins: The diet should ensure adequate provision of vitamins B6, B9, and B12 as cofactors for neurotransmitter synthesis. Deficiencies can lead to mood disorders, anxiety, and cognitive impairment.

Hormonal modulation through diet composition: A low-carbohydrate (high-protein) diet can support longer-lasting satiety through a beneficial effect on ghrelin and PYY profiles.

Environmental enrichment: Systematic use of food puzzles reduces boredom; adjusting the difficulty level to the cat's individual abilities is important. Even if cats do not prefer puzzles, their regular use brings benefits for psychological well-being.

Nutraceuticals: There is preliminary evidence for the effectiveness of antimicrobial peptides, omega-3 acids, and probiotics in alleviating stress and anxiety-related behaviors.

Caregiver education: The necessity of educating caregivers about the behavioral and nutritional needs of cats. Many myths about cats as «low-maintenance» animals lead to the failure to meet their basic needs.

Individualization of nutrition: The observed high inter-individual variability in response to diet indicates the need for individualized nutrition, taking into account the cat's health status, age, activity, and behavioral predispositions.

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